



FIT5057 Project Management Assignment Three

The Stellar Nexus– A Galactica Odyssey

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Contents

DELIVERABLE 1: Agile Project Foundations

Task 1.1 Project Vision and Initial Requirements

Task 1.2: Scrum Persona Development

Task 1.3 Journey Mapping

DELIVERABLE 2: Agile Planning and Sprint Allocation

Task 2.1 Product Backlog

Task 2.2 Strategic Sprint Allocation

Task 2.3 Sprint 1 Execution Plan

DELIVERABLE 3: Agile Reflection and Professional Development

Task 3.1: Reflection on Agile Game

Task 3.2: Personal Reflection and Career Development in Agile Roles

Team Presentation and Agile Artefacts

References

Generative AI - Acknowledgement of Use

Deliverable 1. Agile Project Foundations

Task 1.1: Project Vision and Initial Requirements

Based on Nexaforge', *Stellar Nexus* aims to fill a strategic gap in the game portfolio by developing a unique space exploration RPG with procedurally generated environments and a dynamic diplomacy system.

1. Project Vision

Stellar Nexus fills a strategic gap in NexaForge's game portfolio by creating a space exploration RPG that uses procedural generation and diplomacy-driven gameplay to give players a fresh, dynamic experience. This game aims to deepen player engagement through immersive storytelling and decision-making that shapes the game's universe.

2. Problem Statement

The RPG market lacks space exploration games that balance meaningful choices with player-driven stories. *Stellar Nexus* offers a procedurally generated universe where players can explore, engage with alien civilizations, and make decisions that impact the storyline, allowing for a more personalized experience.

3. High-Level Requirements

The requirements are derived from the project goals and intended user experience:

Requirement	Type	Benefit
Procedural Galaxy Generation	Functional	Offers unique exploration experiences every playthrough. <i>Nguyen, P. (2016)</i>
Diplomacy & Decision-Making System	Functional	Enables impactful choices, affecting relationships with factions. <i>Banerji, A. K. (1979).</i>
In-Game Analytics	Functional	Tracks user interactions, helping to improve player engagement.
High-Quality Visual Assets	Non-Functional	Increases visual immersion, critical for user satisfaction.
Cross-Platform Compatibility	Functional	Expands accessibility, attracting a larger audience. <i>Segun-Falade, O. D., Osundare, O. S., Kedi, W. E., Okeleke, P. A., Ijomah, T. I., & Abdul-Azeez, O. Y. (2024).</i>

Requirement	Type	Benefit
Post-Launch Updates	Non-Functional	Ensures continued satisfaction through bug fixes and new content.

Alignment with Personas

This vision and these requirements were built with two main user personas in mind: *Alex Mercer*, who values deep storytelling and impactful decisions, and *Casey Tan*, a casual gamer focused on exploration and visual appeal. Each feature addresses the unique needs of these players, ensuring that *Stellar Nexus* appeals to both dedicated and casual gamers. *Ghețău, C. (2022).*

Task 1.2: Scrum Persona Development

Persona Name	Description	Goals	Frustrations	Feature Benefit
Alex Mercer 	30, software engineer, enjoys complex RPGs and decision-driven games.	Seeks depth in storytelling and meaningful in-game decisions.	Feels frustrated when decisions don't impact story progression.	The diplomacy system will allow Alex to see his choices impact gameplay.
Casey Tan 	25, casual gamer interested in visuals and space exploration.	Enjoys discovery and wants easy-to-navigate game mechanics.	Overwhelmed by complex controls and repetitive environments.	Procedural generation offers varied exploration without repetition.

Both Alex Mercer and Casey Tan, the personas for *Stellar Nexus*, benefit from distinct game features. Alex, who seeks deep storytelling and meaningful choices, benefits from the diplomacy system, as it lets him see the impact of his decisions, enhancing his sense of control and immersion. For Casey, who enjoys exploration but dislikes complexity, the procedural generation feature provides varied environments without overwhelming controls, ensuring each playthrough feels fresh and engaging. Empathizing with their needs helps create a tailored experience that resonates with both casual and dedicated players

These personas were designed to capture the range of players who might be interested in *Stellar Nexus*, ensuring the game's features remain focused on user needs. Gulliksen, J., Göransson, B., Boivie, I., Blomkvist, S., Persson, J., & Cajander, Å. (2003).

Task 1.3 Journey Mapping

The journey map follows Alex Mercer, a core user persona, as he interacts with the world of *Stellar Nexus*. Alex is a dedicated RPG fan who values meaningful storytelling and decision-making. His journey is broken down into specific stages, highlighting his interactions, frustrations, and areas where improvements can elevate his experience. Lankoski, P. (2011).

Stage	Steps (What is the user doing?)	Touchpoints	Pain Points	Ideal Journey Improvements
Awareness & Consideration	Alex learns about <i>Stellar Nexus</i> through online forums, social media, and reviews.	Game website, online ads, reviews on gaming forums.	Difficulty finding detailed information on gameplay mechanics, unique features, and system requirements.	- More targeted advertising on RPG forums with gameplay highlights. - A comprehensive FAQ section and introductory gameplay videos on the website.
First Interaction	Alex downloads and begins the game, starting with basic tutorials on movement, exploration, and interaction with NPCs.	In-game tutorial, NPC dialogue, introductory UI screens.	Feels overwhelmed by the number of game mechanics introduced early on without sufficient guidance or pacing.	- Gradual, layered tutorials that allow Alex to master one mechanic before introducing the next. - Optional tips available as he progresses for additional guidance.

Stage	Steps (What is the user doing?)	Touchpoints	Pain Points	Ideal Journey Improvements
Engagement & Problem Solving	Alex starts exploring planets, where he encounters an alien faction and is introduced to the diplomacy system.	Decision-making UI, faction-specific dialogues, NPC reactions.	Lacks clarity on how his choices will affect his relationship with factions; unsure of the consequences of each action.	- Visual cues (e.g., icons or colored indicators) showing the impact level of each choice. - Dialogue previews or “hover hints” that explain potential outcomes of actions.
Decision Point	Alex makes a major decision to ally with one faction, impacting his relationship with other factions.	Confirmation dialogues, animated feedback on choice impact, notifications of faction reputation change.	Frustration due to delayed or vague feedback on the immediate effects of his choice, creating uncertainty.	- Real-time feedback animations showing faction reactions. - Clearer indicators of changes in faction reputation post-decision (e.g., reputation bars or loyalty symbols).
Exploration Continuation	Alex continues his journey by exploring other planets, encountering various challenges, quests, and environmental variations.	Exploration UI, HUD with planet info, faction representatives across planets.	Some exploration areas feel repetitive, and unique landmarks are hard to find without clear markers.	- Include unique landmark icons on the map to guide players. - Add varied challenges and mini-events to each biome for a richer experience.

By identifying and addressing these pain points, the game’s UX can be significantly enhanced, creating a smoother journey that aligns with Agile’s user-centered approach.

Deliverable 2. Agile Planning and Sprint Allocation

Task 2.1: Develop a Comprehensive Product Backlog

For *Stellar Nexus*, we'll outline five primary epics and break them down into user stories, applying the INVEST criteria.

Epic	Feature	User Story	Priority (MoSCoW)	Estimated Effort (Story Points)
1. Procedural Galaxy Generation	Planet Generation	As a player, I want a diverse range of planets, so each playthrough feels unique.	Must Have	8
	Hidden Landmarks	As a player, I want rare landmarks (e.g., ancient ruins) to discover, adding mystery to exploration.	Could Have	3
2. Diplomacy System	Faction Decision-Making	As a player, I want my decisions to impact faction relationships, so I feel my choices are meaningful.	Must Have	10
	Reputation Tracking	As a player, I want my reputation with factions to affect how NPCs react to me, increasing immersion.	Should Have	5
3. In-Game Analytics	Player Retention Tracking	As a developer, I want to track user retention, so I can improve engagement strategies.	Should Have	5
	Decision Path Analytics	As a developer, I want to analyze the choices players make, so we can improve decision-making options.	Could Have	4
4. Art and UI/UX Design	Hand-Painted Visuals	As a player, I want visually appealing environments, so I am fully immersed in the game world.	Must Have	6
	Faction-Specific Art Styles	As a player, I want each faction to have a unique art style, making interactions feel distinctive.	Must Have	6

Epic	Feature	User Story	Priority (MoSCoW)	Estimated Effort (Story Points)
5. Post-Launch Bug Fixes	Regular Bug Updates	As a player, I want consistent updates and bug fixes, so I experience smooth gameplay.	Could Have	3
	User Feedback Integration	As a developer, I want to gather and implement player feedback to improve post-launch updates.	Should Have	4

Each epic has been prioritized using the MoSCoW method to ensure that the most critical elements of gameplay are developed first, aligning with project goals and user needs. *AbdElazim, K., Moawad, R., & Elfakharany, E. (2020, February)* ensuring critical features are completed early in development.

Task 2.2: Strategic Sprint Allocation

For *Stellar Nexus*, we have planned five sprints over a 12-week development period. Each sprint focuses on a specific aspect of the game, ensuring that essential features are implemented incrementally, with each sprint delivering functional improvements aligned with the game's vision and player experience.

Sprint	Epic(s) Allocated	Theme	Justification	Trade-offs
Sprint 1	EPIC 1 Procedural Galaxy Generation	Core Exploration Mechanics	The main goal of Sprint 1 is to create a procedural generation framework that generates unique planets with distinct biomes and ecosystems. This foundational system is critical to the exploration-driven nature of <i>Stellar Nexus</i> , allowing players to encounter varied worlds in each playthrough, which increases engagement and supports replay ability.	Delayed visual enhancements (Epic 3), which may impact early visuals in playtests but allows the team to focus on generating functional, varied planetary environments first.
Sprint 2	EPIC 2 Diplomacy System	User Decision-Making	Sprint 2 focuses on implementing the diplomacy mechanics, which allow players to interact meaningfully with various alien factions. By enabling impactful decision-making early in development, we establish core gameplay mechanics that appeal to players who enjoy story-driven choices. This sprint ensures that player actions have visible consequences, deepening player immersion.	Some environmental improvements from Epic 1 might not yet fully reflect player choices, as further design refinements are reserved for later sprints (Epic 3).
Sprint 3	EPIC 3 Art and UI/UX Design	Visual Immersion	This sprint emphasizes the visual quality of the game by enhancing art assets, UI elements, and HUD design. These improvements are essential for creating a visually immersive experience that aligns with the game's high-quality aesthetic goals. Customizable UI options and distinct faction art styles further enrich player experience and appeal to Alex, our core user persona.	Delayed implementation of engagement tracking (Epic 4) means we won't have detailed analytics on player interactions yet, but this ensures visual design is up to standards before collecting user engagement data.

Sprint	Epic(s) Allocated	Theme	Justification	Trade-offs
Sprint 4	EPIC 4 In-Game Analytics	Engagement Tracking	Sprint 4 introduces in-game analytics to track player engagement and decision patterns. By analyzing retention rates, session durations, and decision paths, developers can identify player preferences and pain points. This data-driven approach allows the team to fine-tune gameplay mechanics and identify areas for enhancement, making the game more responsive to player behavior.	Postponing real-time feedback on user behavior means early player feedback relies more on direct playtesting than analytics. However, focusing on core functionality and visuals first allows the analytics to track stable gameplay.
Sprint 5	EPIC 5 Post-Launch Bug Fixes	Stability and User Satisfaction	The final sprint focuses on post-launch support, with bug fixes, quality-of-life improvements, and security updates. By planning for post-launch stability and ongoing updates, the team ensures long-term player satisfaction. Additionally, preparing for future content expansions enables the game to stay engaging over time, addressing the interests of both new and returning players.	Some minor content additions may be postponed until post-launch updates, allowing the team to prioritize stability and bug fixes first to ensure a high-quality launch.

This strategic allocation is intended to deliver a functional product at each stage while addressing key user needs. *Schilling, M. A., & Hill, C. W. (1998).*

Task 2.3: Sprint 1 Execution Plan

Category	Task	Description
Objective	Goal of Sprint 1	The primary goal of Sprint 1 is to implement the Procedural Galaxy Generation feature, which is crucial for creating a unique and varied exploration experience. This feature directly supports the core gameplay experience by ensuring players encounter new planets and environments in every playthrough, fostering replayability and engagement.
Scope	Features and Tasks for Sprint 1: User Story ID: US-01 US-02	The Procedural Galaxy Generation feature focuses on building a framework for creating distinct planets with different ecosystems and attributes. Priority (MoSCoW) US-01 Must Have US-02 Should Have
Definition of Done (DoD)	Criteria to Complete Each User Story	- Code must pass all unit tests and integration tests, ensuring no critical bugs are present. - Playtesting confirms that the generated planets are visually and functionally distinct, providing a unique experience in each playthrough. - Lead developer reviews all code and documentation to ensure quality standards are met. - The procedural generation code is stable and meets the design requirements for performance and variability.
Capacity Alignment	Team Capacity for Sprint 1	- Total Capacity: 20 story points available for Sprint 1. - Allocated Story Points: 18 story points dedicated to critical procedural generation tasks, with a 10% buffer for unexpected issues. - Breakdown: - US-01: 10 story points allocated for creating the foundational procedural generation framework, focusing on the structure and mechanics required to generate varied planets. - US-02: 8 story points dedicated to implementing environmental diversity within the framework, ensuring that each planet has a unique look and feel.

Sprint Review & Retrospective:

Sprint Review: Showcase the procedural galaxy generation feature to stakeholders, with a focus on the unique environments and visual diversity created for planets. Discuss how the variety enhances the player's exploration experience.

Feedback Objectives: Stakeholder feedback will center around the perceived uniqueness and visual appeal of the generated planets, as well as the stability and consistency of the procedural generation framework.

Sprint Retrospective:

- Evaluate the team's collaboration and the efficiency of cross-functional workflows between design and development.
- Identify any technical challenges faced during the sprint, particularly around code stability and the integration of environmental diversity.
- Discuss improvements for coordinating with the art team on asset creation to support future procedural generation needs (e.g., additional textures or biome-specific assets). *Erdoğan, O., Pekkaya, M. E., & Gök, H. (2018).*

Deliverable 3: Agile Reflection and Professional Development

Task 3.1: Reflection on Agile Game (LEGO/Agile Activity Experience)

Beginning: Introduction to the Agile Game Experience

The Agile game, specifically the LEGO activity, was designed to simulate an Agile work environment where each team used LEGO pieces to build a project within limited timeframes. My initial goal for this activity was to understand the practical application of Agile principles, especially iterative development, teamwork, and the role of the Scrum Master in guiding teams. *Keith, C. (2020).*

Middle: Detailed Reflection on the Experience

Agile Principles in Action

The LEGO activity demonstrated key Agile principles such as iteration, collaboration, and flexibility. During each iteration, we built and tested different sections of the LEGO structure, which allowed us to gather immediate feedback and improve our approach. For example, we initially spent too much time perfecting one section, which highlighted the importance of balancing quality with speed and adaptability—a core aspect of Agile’s iterative cycles. This process showed me how Agile’s focus on incremental progress can improve outcomes through continuous feedback and rapid adjustments.

Role of the Scrum Master

The role of the Scrum Master became evident during the activity, as they ensured that the team remained focused, facilitated quick decisions, and helped remove obstacles. Our Scrum Master held brief stand-ups at the start of each iteration, keeping us aligned with our objectives and guiding us in prioritizing tasks. This experience illustrated the Scrum Master’s role in maintaining the team’s momentum and resolving any communication issues, reinforcing my theoretical understanding of how this role supports Agile teams in a dynamic setting.

Challenges and Solutions

One challenge we encountered was miscommunication on task priorities, which initially caused delays. During the retrospective after the first iteration, we discussed these issues and agreed on clearer communication strategies. For example, we implemented a rule where each team member would confirm their tasks aloud to ensure alignment. By adapting our communication method, we reduced misunderstandings and improved our efficiency. This experience reinforced the importance of Agile retrospectives in fostering a culture of continuous improvement.

Conclusion

The Agile LEGO activity provided valuable insights into Agile principles and the role of the Scrum Master. The iterative nature of Agile work and the importance of team cohesion and quick adaptation were evident throughout the activity. This hands-on experience not only solidified my understanding of Agile practices but also highlighted areas for personal growth, such as improving task prioritization and fostering better team communication in future projects.

Task 3.2: Personal Reflection and Career Development in Agile Roles

End: Personal Reflection and Career Development

Aspect	Insight Gained
Role Preference	This experience helped me realize that I enjoy roles that involve organizing and guiding the team, especially roles like Scrum Master. I found that I like helping set priorities, keeping things on track, and encouraging others to work together smoothly.
Skills Developed	Through this activity, I became better at organizing tasks, prioritizing under pressure, and fostering teamwork—skills that are crucial for Agile roles. Taking on some coordination in the activity also gave me valuable experience managing team dynamics.
Future Career Path	The activity strengthened my interest in Agile project management. I plan to work toward an Agile certification, like Certified Scrum Master (CSM), to build on what I learned. I also want to focus on improving my facilitation and conflict resolution skills, which are essential for managing Agile teams.

Team Presentation and Agile Artefacts:



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GENERATIVE AI: Acknowledgement of Use

- AI & Generative AI tools has been SELECTIVELY used within this assessment to make the agile foundation plan close to realistic plans.
- ChatGPT 4.0 has been selective used to rephrase the language of the document to sound more professional and in line with the requirements of Nexaforge Studio.

OpenAI. (2023). *ChatGPT (Mar 14 version) [Large language model]*. <https://chat.openai.com/chat>